HEATMAP NODE · DESIGN DECISIONS REGISTER

HeatMap Node design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/heatmap-node		
Source	docs/06-design-decisions.md @ b9c67bc		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-11-01	Amish Chadha	Kit 1.9.1 upgrade: concept selection (HMN-DDR-004) and the pipeline items O1 to O4 opened as proposed, awaiting Amish; K1, K1b, K2, K3, K4 and K5 recorded as decided under the standing instruction of 2026-10-23; the model rebuilt to the kit 1.9 interface with no design dimension changed
0.2	2026-10-02	Amish Chadha	Amish approved the recommendations for all nine open decisions (HMN-DDR-003 accepted); moved to decisions made
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open decisions, items to confirm, value engineering and decisions made

HeatMap Node design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/`; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

Open decisions

All proposed, awaiting Amish (none decided in the kit 1.9.1 session).

#	Decision	Options	Recommendation	Record
O1	Concept: A (fan-aspirated shield, globe, anemometer) or B (direct natural wet-bulb head); the scores are 140 and 129, within 10 %	A; B; A with a few B reference heads	A, with B heads as references (proposed)	HMN-DDR-004, HMN-OPT-001
O2	Whether to move the sensors toward pedestrian height (PR-009 not met at 2.67 to 2.69 m) or keep 2.8 m for reach and vandal resistance (HMN-DDR-001 D2)	Keep 2.8 m with the correction; lower the arm	Keep, and test 2.0 m against 2.8 m at TRL 4	HMN-PRS-001 PR-009
O3	Raise the budget target or cut cost: the head is USD 6 over the USD 130 target	Raise to USD 136; take the savings listed under Value engineering; leave	Take the savings first; never raise `budget_usd` without Amish	HMN-PRS-001 PR-013
O4	Cross-repo item for FieldNode (STEM S25 fails, fix proposed): stop cell charging above its rated charge temperature (interior 57.2 °C at 50 °C air)	Ask FieldNode to add a cut-off; derate the node	Ask FieldNode	STEM S25, PR-016

Decided under the standing instruction (2026-10-23)

Fixes for failed STEM check rows, made in the kit 1.9.1 session and open to reversal by Amish.

#	Fix	Why	Record
K1	Option B reservoir: fill line at 85 % (15 % headspace), a thin HDPE wall, a gland that is a loose push fit so ice pushes it out, and an invalid-below-5 °C flag on the wet-bulb channel	Water expands 9.07 % on freezing; a brim-full bottle has 3 % headspace and splits at -20 °C	HMN-STM-001 S34, S34b
K1b	Option B reservoir: refill interval of 5 days in hot dry spells, 7 days otherwise, and the wick cord enclosed in a sealed sleeve so only the sensor sleeve evaporates	31.5 g a day evaporates in the worst case; 186 mL usable lasts 5.9 days; an open cord would add about 11 cm2	HMN-STM-001 S31, S31b
K2	The claim about heat island differences restated to 2 °C or more between single readings	Two 0.43 °C random shield errors differ by 0.61 °C in quadrature (1.2 °C at 95 %), and a sunny site reads up to 0.43 °C warm	HMN-STM-001 S20, S20b
K5	Arm check in HMN-CAL-001 restated: minimum yield 110 MPa in place of the typical 145 MPa, and the head's weight added (factor 7.5, was 13.0)	Independent review of HMN-STM-001 (S9); the requirement is still met	HMN-CAL-001 v0.7 H2, H2b
K4	Stale pole base moment of 338 N m in the precis (HMN-PRC-001) corrected to 423 N m, the figure of HMN-CAL-001 H6 after the core moved above the arm	The two documents disagreed; the calculation note governs	HMN-PRC-001 v0.9
K3	Model rebuilt to the kit 1.9 interface at detail level D3 for options A and B; no design dimension changed. New repo-only items: a `zbk` level change for lead B so the two M12 leads do not overlap, and drawn fastener sizes (M3 and M4, nyloc and acorn nuts) that the kit library lacks	The model must be physically constructable	cad/ fidelity.json, REVIEW.md

To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	The globe's top boss is threaded M10 x 1, or an M10 x 1 adapter fits it, and the boss is soldered strongly enough to hang the globe	The hanger tube screws into the boss; the lanyard backs it up	HMN-DDR-003, P7
2	The anemometer's mast is a 16 mm tube that can be drilled for a 5 mm pin, and its cable leaves at the foot of the mast	The mast passes through 16.5 mm holes in the arm and is pinned	HMN-DDR-003, P8
3	The fan's corner holes are on a 50 mm square and take M4 screws	The cowl legs, the fan and the top plate share four screws	HMN-DDR-003, P5
4	The temperature and humidity capsule is 18 mm across and at least 45 mm long	It is a push fit in the 18.5 mm hub	HMN-DDR-003, P6
5	Rubber edge trim for 3 mm sheet with about 1.5 mm of rubber on the bearing edge	Sets where the pole bears in the V	HMN-DDR-003, P1
6	Strap band width (13 mm) and buckle size suit the 16 mm slots, and the band tool gives the 1,000 N preload the clamp check assumes	The twist factor of 2.5 rests on that preload	HMN-CAL-001 [H5]
7	The FieldNode back plate's band slots (51 mm each side of centre) and V-block screw holes (18 mm each side) are as in FND-DWG-101	The adapter V-blocks are drilled and slotted to match	FND-DDR-003; HMN-DDR-003, P12
8	The heat-set inserts' hole size for the printer and filament used	The adapter blocks carry the FieldNode core through them	HMN-DDR-003, P12

Value engineering

Value-engineering target: USD 130 for the sensor head (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 136 (USD 6 over the target). The FieldNode core with its sun shield and the pole adapter (USD 156) are counted against FieldNode; the full node is USD 292, for information. Main cost drivers and savings worth trying:

- The largest lines are the cup anemometer (USD 32), the globe with its hanger tube (USD 24), the shield (USD 15), the temperature and humidity sensor (USD 14) and the harness (USD 12); these five are 71 % of the head.
- Making the design constructable added USD 9: cheeks, wider sheet and trim on the saddle (USD 3), the M10 hanger tube and nuts (USD 2), a fourth rod and two spacers (USD 1), and bolts, sleeves, a plug and long cable ties (USD 3).
- Savings worth trying: one 8-pin lead in place of two 5-pin leads, if FieldNode's port design allows it (about USD 4); a float ball bought with a threaded boss already fitted, avoiding an adapter; printing the cheeks in ASA is not recommended, since they carry the arm.

Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 items D1 to D8: budget covers the sensor head only; arm at about 2.8 m with a height correction; 150 mm globe; cup anemometer; WBGT derived by the Liljegren method; FieldNode core with TwinKit or a public LoRaWAN network; open data, environmental channels only (D5, passive shield, later superseded by N2)	Amish: "i accept all your recommendations, go with them across all repos."	HMN-DDR-001, HMN-DDR-002
2026-09-25	N1: pole adapter counted against FieldNode and a formed 3 mm sheet saddle	Amish, same instruction	HMN-DDR-002
2026-09-25	N2: fan-aspirated shield, run 6 s before each reading every 3 min	Amish, same instruction	HMN-DDR-002
2026-09-25	N3: R15 mass limit applies to the sensor head only	Amish, same instruction	HMN-DDR-002
2026-09-25	N4: R8 at +50 °C and FieldNode's hot-climate sun shield on HeatMap nodes	Amish, same instruction	HMN-DDR-002
2026-09-25	N5: arm toward the equator with the FieldNode core below it	Amish, same instruction	HMN-DDR-002
2026-09-25	N6: R4 restated to 0.8 to 20 m/s with calm intervals flagged	Amish, same instruction	HMN-DDR-002
2026-09-26	N7: `budget_usd` raised from $120 to $130	Amish: "I am ok with the budget top ups"	HMN-DDR-002 v0.2
2026-10-01	`budget_usd` is a hypothetical value-engineering target, not a limit; cost is reported against it	Amish: "the budgets are a hypothethical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	This register; HMN-CAL-001 v0.4
2026-10-02	Design for construction accepted: the thirteen changes of HMN-DDR-003 as made, including the arm end cap of the 2026-09-26 appearance review (P9) (open item 1)	Amish: "i approve your recommendations for all 555 open decisions."	HMN-DDR-003, Table 1
2026-10-02	Sensor hub kept on three spokes (option a) and included in the TRL 4 side-by-side shield test (open item 2)	Amish: "i approve your recommendations for all 555 open decisions."	HMN-DDR-003, A1
2026-10-02	FieldNode core mounted above the arm (option a), provided its lid stays reachable from the ladder or lift used to fit the arm; if not, the core goes on the pole's east or west face (option c) (open item 3)	Amish: "i approve your recommendations for all 555 open decisions."	HMN-DDR-002, N8; HMN-CAL-001 [E3]
2026-10-02	First partner and city: a hot US city with a dedicated heat office; first candidate to approach		HMN-DDR-001, O1

Date	Decision	Decided by	Record
	the City of Phoenix Office of Heat Response and Mitigation, using the US915 band (open item 4)	Amish: "i approve your recommendations for all 555 open decisions."	
2026-10-02	Data routed through a TwinKit gateway and published from CityTwin's open data export under an open licence, with Amish's lab as publisher until the partner city takes it over; mirrored on the city's own open data portal if it has one (open item 5)	Amish: "i approve your recommendations for all 555 open decisions."	HMN-DDR-001, O2
2026-10-02	Sensor ports follow FieldNode's candidate pinout: port A, the I2C sensor on data pins 2 and 4 with sensor and fan on the switched rail; port B, the thermistor on analog pin 5 and the anemometer pulse on pin 2 (open item 6)	Amish: "i approve your recommendations for all 555 open decisions."	FieldNode FND-DDR-001, O2; HMN-DDR-002 cross-repo actions
2026-10-02	FieldNode internals kept as stand-ins in the appearance model; the FieldNode repo is asked to confirm the cell format and board position at its next update (open item 7)	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW.md, 2026-09-26
2026-10-02	Name plate kept on the FieldNode sun shield; side vent slots dropped unless FieldNode adopts them (open item 8)	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW.md, 2026-09-26
2026-10-02	Raised seam ring at the globe equator accepted for appearance (open item 9)	Amish: "i approve your recommendations for all 555 open decisions."	REVIEW.md, 2026-09-26